

Architectural Particular Specification

TIMBER DOORS

SECTION 7.2

Rev	Date	Description
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SECTION 7.2

TIMBER DOORS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The scope of works includes, but is not limited to, the supply and installation of timber doors and frames (non-fire rated and fire rated), in accordance with the contract documents and this particular specification including the followings:
1. All fire rated and non-rated timber doors & frames.
 2. Plastic Laminate Face
 3. Vision Panel
 4. Beads, trim, seals and all other associated accessories
 5. Fire test and assessment of fire-rated door assemblies.
 6. Wood Veneer
- B. Where required to achieve fire resistance integrity to meet all latest statutory requirements, including the design, fabrication, installation, testing and statutory certification necessary for the completion of the system and shall incorporate all associated works in the application of the building finishes to suit, the earthing requirements, and sealing around the installations, herein called the System.
- C. For all fire rated doors, apart from this specification, the contractor shall also refer to and comply with the latest prescriptions in the Code of Practice for Fire Safety in Building 2011 and all its corrigenda.
- D. Acoustic rated solid core timber door according to Specification of Acoustic Performance of Entry Doors under this Contract.

1.02 REFERENCES

- A. British Standards
1. BS 476: Part 20: 1987 (1990) - Method of Determination of the Fire Resistance of Elements of Construction.
 2. BS 476 :Part 22 : 1987 - Method of Determination of the Fire Resistance of Non-loadbearing Elements of Construction.
 3. BS 476: Part 23: 1987 - Method of Determination of the Contribution of Components to the Fire Resistance of a Structure.
 4. BS 476: Section 31.1: 1983 - Method of Measuring Smoke Penetration Through Doorsets and Shutter Assemblies Under Ambient Temperature Conditions.
 5. BS 1142: 1989 - Specification for Fibre Building Boards.6. BS 1186: Part 1: 1991 - Specification for Timber
 6. BS 1186: Part 2: 1988 - Specification for Workmanship of Timber and Joinery
 7. BS 1245: 1975 - Metal Door Frames (Steel).
 8. BS 5872 : 1980 – Specification for Locks and Latches for Doors in Buildings
 9. BS 8214: 1990 - Code of Practice for Fire Door Assemblies with Non-Metallic leaves.

10. BS 8220: Part 2: 1995 - Guide to Security of Building Against Crime: Offices and shops.
11. BS DD 171: 1987 - Guide to Specifying Performance Requirements For Hinged or Pivoted Doors.
12. PD 6512: Part 1: 1985 - Use of Elements of Structural Fire Protection with Particular Reference to the Recommendation Given In BS 5588: Guide to Fire Doors

B. American Standards

1. ANSI/NFPA 252-1990: Fire Tests of Door Assemblies.
2. ANSI/NFPA 255-1990: Surface Burning Characteristics of Building Materials.
3. ANSI/NWWDA 1.S.1 - 87 : Wood Flush Doors
4. UL 1785 - 95 : Air Leakage Tests for Door Assemblies
5. ASTM E152 - 81a : Method of Fire Tests of Door Assemblies (Hose Stream Test)
6. ASTM E283-73: Door Air Infiltration Test
7. ASTM E283-84: Test methods of Air Leakage Through Exterior Windows, Curtain Wall and Doors.
8. American Wood Work Institute: AWI Quality Standard for Architectural Flush Doors.
9. NFPA 105-89: Installation of Smoke Control Door Assemblies
10. NFPA 80: Fire Doors and Windows (Current Edition)

1.03 QUALITY ASSURANCE

- A. Qualification of Timber Door Manufacturer: Company specializing in the manufacturing of high-quality, commercial-grade, fire rated timber doors with minimum of ten (10) years documented experience (most recent five (5) years prior to Tender date).
- B. Acoustic doors shall be installed by specialist installer having at least 10 years acoustic door installation experience.

1.04 PERFORMANCE REQUIREMENTS

- A. Doors Generally

1. Bottom edges of door shall be primed before hanging.
2. Sample of a finished door must be submitted to the Architect for confirmation of no adverse comment prior to installation.
3. Doors with fire rated requirements shall be provided with fire certificates on testing as recognized by Independent Checker (for compliance of Buildings Ordinance and Regulations requirements) and Fire Services Department.
4. Bottom edges of door shall be primed before hanging.
5. Sample of a finished door must be submitted to the Architect for confirmation of no adverse comment prior to installation.
6. Doors with fire rated requirements shall be provided with fire certificates on testing as recognized by Independent Checker (for compliance of Buildings Ordinance and Regulations requirements) and Fire Services Department.
7. Acoustic performance of acoustic doors with weighted acoustic performance 35dB Rw or higher to be independently tested and certified to BS EN ISO 140 Part 3.
8. Contractor shall submit the latest certifications to demonstrate compliances to the above performance standards.

9. Timber shall be to BS1186 Part 1 and in particular timber for cores and for lippings and shall be of sustainable species of hardwood.
10. Door leaves over 2230mm/H shall use 4 nos. of stainless steel ball bearing butt hinges.
11. Frame and architrave materials shall be hardwood material. Dimension shall refer construction details respectively. Frame is made by one piece type solid timber. Frame head is made by mitre joint at corners. Sub-frame 130mmL meranti, is fitted on wall first before installation of main frame which is coated with preservative oil. Frame size over 150mm length will use 2 pieces of same material combined together. Sub-frame will increase the size as main frame.
12. All doors shall have a label attached to the top which identifies it in relation to the door schedules, its position in the works and its fire rating.

B. Timber Door and Frame Quality:

1. Timber used for the construction of wood doors shall be kiln-dried materials at a moisture content of 10-16% to suit intended conditions of use. They shall be free of splits, shakes and checks, even on concealed or door core timbers. Timber material shall be uniform; timber with marked changes between heartwood and sapwood shall not be used.
2. All timber with unsound knots shall be plugged and filled with inorganic fillers, even on concealed or door core timbers. Exposed timber door surfaces shall not have any knots.
3. For internal timber door joinery, up to 4 growth rings per 25 mm is the acceptable maximum. For external wood door joinery, 6 growth rings per 25 mm is the maximum limit.
4. Door frame timbers shall have a grain slope better than 1 in 15.
5. Cutting of timber door for glazing and other apertures shall not remove any part of any structural members of the door leaf. Internal stiles and rails constitute door structural members.
6. All timber doors shall be manufactured to BS1186: Part 2 and AWI Quality Standards for Architectural Flush Doors: Premium Grade Construction.

C. Fire & Smoke Protection: Quality standard for architectural flush doors: Premium Grade Construction.

1. All fire doors shall have the ability to resist burn-through and remain free from critical levels of distortion under fire exposure.
2. All fire rated doors shall be tested be certified by a HOKLAS recognised independent laboratory as being capable of providing fire resistance periods as specified to the specified fire rating requirements and in accordance with BS 476 part 20 & 22. Fire certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals. Doors shall be periodically checked and inspected by the fire testing laboratories to ensure that they comply with those which have been fire tested. Fire rated doors for each type of door shall be tested for fire performance and the fire test report shall be submitted to the Architect for vetting.
3. Fire rated doors shall be certified for single or double-acting, single or double-leaf, latched and unlatched door configurations in a BS 476 Part 22 fire test. All doors shall be fire tested and approved with the specified door seals, ironmongery, vision panels and door frames together as an integral assembly. Where fire doors are to be fitted in non-masonry constructions, which would be expected to affect fire ratings, such as plasterboard partitions, provide fire test data specific to such constructions.

4. All fire-resisting door assembly with the necessary ironmongery fitted to have the minimum period attained when tested for integrity to B.S.476, part 22, 1987 and as per Code of Practice for Fire Resisting Construction.
5. Provide evidence that the fire-rated door assemblies with the necessary ironmongery fitted fully comply with Ordinance/Regulation of Hong Kong.
6. Doorsets shall provide effective smoke stops in case of fire. Smoke control to BS 476 Part 31.1 (3m³ per lin.m per hour @25Pa).
7. No fire resisting doors shall be hung on butts made of combustible materials or metals, with melting point less than 800°C.
8. Where fire doors are to be fitted in non-masonry constructions, which would be expected to affect fire ratings, such as plasterboard partitions, provide fire test data specific to such constructions.
9. All door frame size and door leaf thickness for FRP doors are for indication showing the design intent and minimum dimension required. The contractor shall be responsible to design and supply the doorset in compliance with the relevant B.S. standard and statutory regulations without any additional cost claim.
10. Any voids due to over mortising as a result of fitting ironmongery shall be filled with suitable weather proof intumescent fire stop material. All gaps or voids around door edge lippings, around apertures and within core material shall be filled with suitable weather proof intumescent fire stop paste.
11. All fire doors shall meet the stability, integrity and insulation requirements of BS 476: Parts 20 and 22 ANSI/NFPA 252 and 253. All doors shall be fire tested and approved with the specified door seals, ironmongery and timber door together as an integral assembly. Any deviations in the door and frame construction, material, ironmongery type, and method of fixing/installation shall be assessed by a recognized Testing Agency.
12. All the fire resisting doorsets should have FRP certificate complying with Buildings Ordinance and Regulations requirements and Fire Services Department requirement and BS 476 Part 20 to 24 : 1987.
13. All the fire resisting enclosure should be constructed of a material which can achieve the required FRP and tested in accordance with BS 476 Part 20-24. Insulation is also required to comply with BS476 Part 8.
14. The Contractor shall procure from the manufacturer, the necessary fire certificates for the installed System. The Contractor shall note that acceptance of the works shall be subject to obtaining the necessary certificate and the satisfactory statutory inspections of the entire System, and the satisfactory inspection of the maintenance contractor appointed by the Employer.
15. All door fire seal, smoke seal and weather strippings shall comply with the following requirements:
 - i) Fire Test Certificate and Approval
 - ii) All assemblies of fire rated doors and frames are to be identified by the SISIR label.
 - iii) All door frames materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL), material capable of passing ASTM E152-81a, ASTM E283-73, NFPA 105-89, NFPA 80, and UL 1784.

- iv) Alternatively, all firestop and barrier seal system materials shall be certified by the British Board of Agreement (BBA), tested to BS 476 : Part 20 and Part 22, and all referenced Standards cited herein.
- v) All door fire seals and smoke seals shall be "CERTIFIRE" and BBA approved, or equivalent UL-labeled seals.
- vi) When tested to BS 476 : Section 31.1, air infiltration/exfiltration shall not exceed 7m³/h or 0.25m³/h per metre of any door perimeter air sealing, whichever is greater, at pressure of 50Pa above atmospheric pressure.
- vii) All door smoke seals shall be tested to BS 476 : Section 31.3 and NFPA 105-89 to restrict the passage of hot smoke to a maximum 6% obscuration per foot (19% obscuration per metre).

D. Acoustical Performance

1. Acoustic timber doors shall be installed as indicated on the drawings with specified minimum STC rating as described in Section 08 13 00.
2. All acoustical doors shall be installed with perimeter seals meeting test requirements of ASTM E90-90 and ASTM E1408-91.

E. Physical Endurance Performance Requirements

1. All timber doors shall be Grade II Heavy-Duty Level capable of meeting the door tests of BS DD 171: 1987 for impact, torsion and closing against an obstruction tests.
2. All door frames shall be Grade II Heavy-Duty Level capable of meeting the door tests of ANSI A250.4-1994 with 1 million swing operating cycles and twist tested 23 times at regular intervals during the cycle test.
3. After completion of BS DD 171 test, no structural failure warping exceeding 3 mm, or deformation at hardware provisions cut outs shall occur. Submit test results to the Project Manager for review.
4. Jointing and fixing of all timber door joinery be such that shrinkage in any direction shall be accommodated in the joints, and shall not impair the strength and/or appearance of any timber door. Should any shrinkage or warping of joinery take place so as to impair the strength and/or appearance of the timber door during the Defect Liability Period, the Contractor shall make good all defects to the satisfaction of the Architect. Defective timber doors shall be taken down and replaced. Any Work disturbed shall be made good at the expense of the Contractor.
5. All door frames and frame anchors shall be tested and meet the requirement of ANSI A250.5-1994.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITION

- A. The Contractor and Door Manufacturer shall be aware of the Indoor psychometric perimeters used by the Consulting Mechanical Engineer in determining vapour barrier, insulation, indoor temperature and relative humidity requirements, heating and cooling loads.

- B. The Contractor, together with the Timber Door Manufacturer shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and indoor psychometric conditions encountered without warpage, face de-lamination, material deterioration and other defect.

1.07 SUBMITTALS

- A. Samples:
1. Provide 600 mm x 600 mm size samples of part of each door panel showing core construction, edge lipping, face materials, frame and vision panel.
 2. Provide a 600 x 600 sample of the door and door frame assembly with hinges, lock, handles and all required smoke seals fitted and installed.
- B. Shop Drawings:
1. Indicate profiles, sizes, connection attachments, reinforcing, anchorage, core and edge construction.
 2. Include Sections and elevations in 1:20 scale and details in 1:2 scale.
 3. Indicate structural opening dimensions and door opening dimensions, locations, length & type of all intumescent and smoke seals and cut outs. Show details.
 4. Indicate surface weight of door leaf including all accessories and ironmongery installed to achieve the specified door acoustic rating.
- C. Computer generated door schedules and shop drawings fully detailing the requirements of each door assembly in sufficient detail to ensure correct manufacture.
- D. Computer generated door-by-door "set type" ironmongery schedules, each set fully detailed and listing each door to which it applies, cross-referenced to an index listing each door and its proposed set type and total quantities.
- E. Individual, independent fire test/assessment certificates, specific to each fire rated door assembly proposed for the Works, confirming compliance with required ratings.
- F. Independent smoke test certificates, confirming compliance of proposed smoke control assemblies with required ratings.
- G. Catalogue of proposed acoustic door seals.
- H. Acoustic test certificates to demonstrate compliance with the specified acoustic rating for the proposed acoustic doors. Where valid acoustic test certificates are not available, the contractor shall, at their own cost, conduct necessary acoustic tests to demonstrate compliance. All acoustic tests shall be performed by an independent recognised acoustic laboratory to be approved by the Architect.
- I. Any further information required by the Architect and other certificates required therein.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE AND HANDLING

- A. Deliver and store on project Site all timber doors and frames only when the Project Building is completely enclosed, moisture-laden materials such as concrete, plaster, etc. have largely be dry, and heating or cooling system are functioning to maintain uniform indoor psychometric conditions.

- B. Store all materials and items away from windows or exposed openings. They shall not be exposed to sunlight or subject to wetting. Veneered doors shall be covered so that no part of the face of any door is exposed to light which will cause fading or discoloration.
- C. Protect all timber doors from damage, staining, traffic and Work of other trades. Ensure that all door edges and corners are protected from chipping or other damage.
- D. Door leaves shall be stored flat on leveled 100 x 50 wood sleepers on length equal to full width of doors. Support sleepers shall be covered with clean cardboard to prevent marking of the bottom door.
- E. Check all deliveries and consignments for proper quantity, wood species, and grading. Ensure mill certificates and fire-retardant treatment certificates accompany all deliveries and consignments. Ensure that all fire doors are affixed with Fire test/approval certificate metal labels, located in a position acceptable to the Architect.
- F. Never stand panels on end against a wall for any length of time. If it is not convenient to lay them flat, they shall stand on their sides with care taken not to chip the sides and edges. Protect all edges and corners from damage.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

- A. All assemblies of fire rated doors and frames shall have test certificate(s) to demonstrate the whole installation are acceptable to the Buildings Department (BD) and Fire Service Department (FSD), with regard to the specified Fire Rating Period.
- B. All door frames materials shall have been tested, approved and listed by U.S. Underwriter's Laboratory (UL), material capable of passing ASTM E152-81a, ASTM E283-73, NFPA 105-89, NFPA 80, and UL 1784.
- C. Alternatively, all firestop and barrier seal system materials shall be certified by the British Board of Agrément (BBA), tested to BS 476: Part 20 and Part 22, and all referenced Standards cited herein.
- D. All door fire seals and smoke seals shall be "CERTIFIRE" and BBA approved, or equivalent UL-labeled seals.
- E. When tested to BS 476 : Section 31.1, air infiltration/exfiltration shall not exceed 7m3/h or 0.25m3/h per metre of any door perimeter air sealing, whichever is greater, at pressure of 50Pa above atmospheric pressure.
- F. All door smoke seals shall be tested to BS 476 : Section 31.3 and NFPA 105-89 to restrict the passage of hot smoke to a maximum 6% obscuration per foot (19% obscuration per metre).

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 MATERIALS

A. Door: Comprising:

1. Laminated Softwood Batten
 - a. Western red cedar battens glued together vertically to form door solid core. The battens shall be laid with grain in alternate and opposite direction to balance stress and thus reduce distortion.
 - b. Minimum thickness of battens shall be 44 mm. Minimum density of Western Red Cedar shall be 420 kg/m³.
 - c. Provide 12 mm thick lipping to all four sides. Lippings formed from hardwood of a minimum density of 650 kg/m³.
 - d. Use Type WPB resorcinol formaldehyde adhesive conforming to BS 1204 and BS 5442: Part 3 for gluing between battens and for bonding core with face material.
 - e. Fixing of lipping to be either mechanically fixed or by use of adhesives cured by radio frequency.
 - f. Overall, net of finishes door leaf thickness shall be 44 mm thick minimum, 36kg./m² minimum approximate weight. Flush door construction.
2. Mineral Core
 - a. Mineral core of aerated magnesium oxychloride cast in density of 700kg/m³ and 38mm thickness. Install hardwood blockings within door core for installation of all ironmongery. Hardwood blockings shall have same density as mineral core.
 - b. Hardwood edge lippings shall be applied to all 4 perimeter edge of door core as indicated above.
 - c. Overall door leaf thickness shall be 44 mm minimum thick, 38 kg/m² minimum approximate weight
3. All fire-rated wood doors shall be fire-tested and certified to rating up to 240/240/240 FRR. All fire-rated wood doors shall have non-removable fire labels mounted on the top of the door.
4. Doors for Security Control Room, Management Offices, Network Operation Centre, Office and other specified areas, shall have a minimum STC rating of 35, installed and be installed with fully rebated acoustical seals.
5. Provide fire doors with fully concealed intumescent seals.

B. Door Frame: Comprising:

1. Frames to be provided together with doors as one complete door sets.
2. Door frame jamb depth, profile and rebate vary. Refer to the Drawings for each location and application. 30mm minimum door frame stop for fire-rated doors.
3. All door hinge fixing at exterior and high security doors shall be counter sunk screwed to the door frame.
4. Door sub-frame to be installed and provided.
5. Frames finish refer to drawing.

6. Frames are San Cheong with a minimum density of 640kg/m³ and nutricellulose finish painting are coated matching laminate color.
7. For veneered door, 2 coats of semi-matt clear lacquer of 30% gloss are coated. Frame dimension is 47x150mm with stop 14mm and one piece type.
8. Alternative, the material can be stainless steel of grade 316 satin finish. Cement sand grouting is filled into steel frame which is done by the contractor. All frame dimensions have to follow fire assessments.

2.02 ANCILLARY MATERIALS

A. Fixing Anchors :

1. Three adjustable door fixing shall be supplied for each jamb of all door frames. "T"-shape fixing anchors shall be 1.2mm thick minimum corrugated galvanised steel.
2. Adjustable L-shape floor anchors shall be 2mm thick, galvanised steel. One base anchor per jamb.

B. Fixing anchors for face fixed frames.

1. Provide a minimum of three number adjustable jamb anchors per jamb. Provide suitable fixing bolts for installation into reinforced concrete. Fixing bolts to be countersunk into frame and conceded with a face cap. Face cap to be fabricated from metal, finished to match the frame.
2. Rubber Silencers : 10mm diameter by 2mm thick rubber silencers fixed to frame rebate. Install 3 silencers on lock jamb side for single doors. Place one silencer adjacent to lock strike plate and the other two approximately 450mm from the first.

For double doors, install one silencer for each door leaf at head spaced approximately 150mm from meeting joint between the door leaves.

C. Glazing Bead :

Square bead type flush with door faces, hardwood species to match lipping.

D. Glass :

Glazing shall be clear laminated glass for non-rated doors and non-wired insulated clear tempered glass for fire rated doors. Glazing detail and glass type shall comply with the specified fire resistant rating and/or acoustic rating. All vision panels shall be factory prepared and installed

E. Vision panels :

Splayed and square beads type with same hardwood species to match lipping. Glazing shall be clear laminated glass for non-rated doors and non-wired insulated clear tempered glass for fire rated doors. Glazing detail and glass type shall comply with the specified fire resistant rating and/or acoustic rating. All beadings are coated nutricellulose paint or 2 coats of semi- matt clear lacquer.

F. Subframe :

18mm film plywood which is coated with preservative oil. This is provided and installed by the contractor.

G. Signage/Kick Plate:

Signage and kick plate shall be fixed flushed with the door surface in factory prior to delivery to site.

- H. Adhesive :
Type 1 PVA adhesive
- I. Plastic Laminate :
1mm thick or equal approved. Good grade veneers are selected which thickness is 0.6mm . The grain pattern is crown cut and surface with 2 coats of semi-matt clear lacquer.
- J. Acoustic seals shall be manufactured by Raven or equivalent to achieve STC 35.and approved by the Architect. (Ref: Lorient Le 1212 and RP8).
- K. Refer to door schedule for locations of doors with acoustic seals.
- L. Intumescent Fire and Smoke Seals: Intumescent fire and smoke seals to prevent penetration by hot gases, smoke, heat and flame. They shall comply with NFPA-105 (1989), BS 476 : Part 20, 23 & 31.1, and ASTM E153-81a. Intumescent Fire and smoke seal types and locations shall be selected to meet specific fire-resistance requirement of each door and test certificate requirements.
- M. Protective Film: Strippable protective polyvinyl chloride (PVC) film of 0.3mm thick to cover all door frames to prevent damage during transport and installation.

2.03 FABRICATION

- A. Fabricated all timber doors sets to AWI Quality Standard Premium Construction Grade. All timber doors shall be rigidly built and free from warp or buckle and any defects or damages.
- B. Select core materials as required to meet the required fire or acoustic performance, or manufacturer's equivalent, subject to Architects approval.
- C. Neatly route prepare and install all wood doors to receive ironmongery, intumescent and smoke seals, and weatherstripping provided under Section 087100.
- D. Prime and finish all cut-outs, recesses and grooves for door ironmongery, intumescent and smoke seals, and weatherstripping. The colour of the finish is to match the laminate colour.
- E. Factory finish exposed lipping with two coats of alkyd enamel paint, semigloss finish, spray applied, colour to match laminate, for laminate faced doors.
- F. For lacquer finish doors, provide lipping quarter sawn, finished to match face of door, specified under section 06 40 00 work.
- G. Under cut non-fire rated doors as indicated on the Drawings. Install door bottom hardwood lipping at all under cut doors. Ensure weatherproof installation. Fine sand and pre-finish all wood doors with a factory applied seal, or primer for doors to be painted.

All door ironmongery and hardware shall be fully rebated into door. Surface mounted ironmongery or other hardware will not be permitted.

2.04 FINISHES/FINISHING TREATMENTS

A. Finishing and Protecting

1. Sand all joinery to give smooth, flat surfaces suitable to receive specified finishes.
2. Protect completed joinery against damage, dirt, moisture and other deleterious substances.
3. Do not deliver joinery to site until immediately before required for fixing. Keep clean and dry and store in conditions of temperature and humidity similar to those prevail in the completed building.
4. Do not fix joinery before building is weather tight, wet trades have finished their work and the building is well dried out.
5. Before installation, ensure that the adjacent construction and relative humidity of the base are suitable.

B. Laminated Plastic Veneers

1. Apply sheets in accordance with manufacturer's recommendation.
2. Condition sheets before bonding. When not otherwise specified, apply to the reverse side of flat boards a balancing veneer of similar construction to the decorative veneer and from the same manufacturer.
3. Bond in presses whenever possible.
4. Finished components to be free from bow, twist, scratches, chipping, cracks, pipping, depressions, glue spill, staining, defects in colour and pattern and the like.
5. All joints exposed to view in the finished work to be tight butted and true with no lipping. Chamfer edges at all external angles.
6. Do not fix laminated plastic sheet to timber with moisture content of more than 16%.
7. The Standard of laminate must meet the GREENGuard Indoor Air Quality Certificate.
8. Doors to be factory-finished comprising the laminate, and the finish to the lipping. For the plastic laminate, it should meet the heavy use requirements of FIRA 6250 on horizontal surfaces further described under the relevant section of the PS.A 23 for Plastic Laminate.

C. Wood Veneers

1. Condition core material and veneers before bonding. When not otherwise specified, apply to the reverse side of flat boards a balancing veneer with the same moisture and temperature movement characteristics as the facing veneer.
2. Set out veneers so that features and pattern are aligned and either "book-matched" or "slip-matched" of minimum width per strip not less than 125mm running a minimum full length of 2500mm. Treat veneers finishes with approved non-yellowing, non-oxidizing agent compatible with the adhesive used. Apply veneers with edges tight butted, no gaps or other open defects and no lipping.
3. Bond in presses whenever possible.
4. Finished components to be free from bow, twist, scratches, chipping, pipping, depressions, glue spill, staining and the like.
5. Sand to a fine, smooth finish free from sanding marks.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify substrate conditions, opening sizes and tolerances are acceptable for proper installation.
- B. Verify that door frame has been properly installed to receive door.

3.02 INSTALLATION

- A. Allow adequate time for all doors to condition and acclimatise to their psychometric environment prior to installation.
- B. Install doors in accordance with fire rating approval requirements and Manufacturer's instructions.
- C. Coordinate installation of doors with installation of ironmongery, and weatherstrippings, smoke seals, and intumescent.
- D. Doors to be finished ready to receive finish painting.
- E. Set frame within opening, plumb jambs and level head. Brace frame as required. Build frame into wall. Fully fill the frame with mortar as the work progresses. The frame alignment is to be checked as the wall is built.
- F. Coordinate with adjacent wall construction for anchor placement.
 - 1. Provide mortar filling of door frames at all fire-rated doors and at locations indicated on the Drawings.
 - 2. The voids between door frames and wall shall be carefully packed back and made good with proper caulking, sealant and plaster.
- G. All frames and architraves shall be fully sealed to the structure with mortar or plaster. Gaps not exceeding 6mm may alternatively be sealed with a non-setting polysulphide or silicone sealant.
- H. No light shall be visible through any installed doorsets when observed in darkness from either side, with room lighting on the opposite side. Where specified in the Drawings or Schedules, doors shall be fitted with an automatic door closer to eliminate slamming noise.
- I. Where specified in the Drawings or Schedules, compression seals shall be fitted to the door rebate sides and head, and shall meet the door leaf at all points when the door is closed.
- J. Seals shall not be stretched.
- K. The door seals shall not inhibit normal use of the door or make the door difficult to close.
- L. Where specified in the Drawings or Schedules, perimeter seals and bottom drop seals shall be provided as a standard integrated fixture of the door assembly. A drop type threshold seal shall be rebated in to the bottom of the door. This shall be activated by door closure.

- M. Intumescent fire seals shall be provided in addition if required, as specified by the Architect.
- N. Each individually installed door set shall provide the specified minimum acoustic sound ratings, as measured in accordance with the latest revisions of ASTM E90-85 and ASTM E413-87, or their equivalent.
- O. Acoustic doors shall be installed in general accordance with the following installation sequence.
1. Check and confirm the sizes of structural opening and door opening against the door schedule.
 2. The voids between door frames and wall shall be carefully packed back and made good with proper caulking, sealant and plaster.
 3. Fix the door frame in place and leave all fixings loose.
 4. The door leaf shall now be hung and adjusted such that the seals on the hinge post side are correctly functioning. With the door closed, the frame should be adjusted up to the door blank, ensuring that there is no undue strain between the blank and the frame. The gap at the top of the door shall be uniform and parallel, achieved if necessary by packing up the threshold.
 5. When satisfied with the adjustment, the frame should now be fixed in place, both horizontally and vertically, using packing pieces where necessary, so as not to destroy the regular opening between the frame and the leaf.
 6. The striker plate can now be adjusted away from the door blank such that there is sufficient clearance to avoid rubbing when the door is opened or closed.
 7. Where appropriate, the threshold can now be packed up and fixed, ensuring that there is no strain or bow introduced and the latch plate must now receive its final adjustment such that the door compresses the seals evenly around the perimeter.
 8. All frames and architraves shall then be fully sealed to the structure with mortar or plaster.
 9. Notwithstanding the guidance on materials and fixings given above, the Contractor must ensure full compliance with the required acoustic ratings given in the Schedule in Section 081300.
- P. Seals and sealing mechanisms shall be adequately protected against damage during transit and installation. Site cutting of seals shall be limited to an absolute minimum, and shall be approved by the Architect.
- Q. Threshold sealing mechanisms shall be adjustable, to allow for the accommodation of door movement and normal building tolerance without loss of performance, but the seals shall not be used to take up tolerances greater than +/- 30% of its design dimension without being overstretched. The threshold seal shall consist of a compression seal designed for use with an associated threshold limit. Compression seals shall be fitted to the frames of all doors to form continuous seals around the jambs and heads.

R. Fixing

1. Fixing Centers For Timber Frames :

Door frames shall be pre-drilled off site for fixing to the surrounding structure. Fixing holes shall be covered in matching colour glommets or by intumescent strippings of fire rated openings. When not predrilled or specified otherwise, position fixings 150mm from each end of jamb, adjacent to each hanging point and at 600mm maximum centres. To further check with the contractor on fixing detail of door..

2. Prepared Openings :

Ensure that lintel, block-work opening, or concrete wall opening are properly constructed. Refer also to the section on concrete lintel.

3. Building In :

4. Components which are being built in must be braced and protected as necessary to prevent distortion and damage during erection of adjacent structure.

5. Pointing :

Ensure joints are dry and free from dust and grease. Mask adjacent surfaces which would be impossible to clean if smeared with sealant.

Prepare joint and apply vinyl based sealant to manufacturer's recommendations forming an approved profile.

Remove excess sealant and leave clean.

6. Backing Strips :

Insert in all joints to be pointed with sealant, unless shown otherwise. Do not leave gaps and do not reduce depth of recess for sealant to less than depth recommended by manufacturer.

7. Ironmongery:

Assemble and fix carefully and accurately using fastenings with matching finish supplied by ironmongery manufacturer. Prevent damage to ironmongery and adjacent surfaces. At completion check, adjust and lubricate as necessary to ensure correct functioning.

S. Adjusting:

Test for smooth operation through full range of door swing; make necessary adjustments.

3.03 CO-ORDINATION

- A. Coordinate adjustment of doors with installation of ironmongery. Adjust doors and ironmongery for smooth and balanced door movement.

3.04 CLEANING

- A. Clean all wood doors free from dirt, dust and stains. Doors shall neither squeak nor rattle when operated.

3.05 PROTECTION

- A. Protect all doors during and after installation until Project Handover, with heavy duty protection material.
- B. Protect all doors from the Work of other trades, traffic and weather to prevent damage, warpage, staining, etc. Additional protection shall be provided by the Contractor at heavily traffic areas and locations subject to impact above.
- C. The Contractor shall repair, or if necessary replace, all damaged wood doors and accessories to the satisfaction of the Architect.
- D. The Contractor shall:-
 - 1. Provide suitable, secure, clean, dry, weatherproof stores and receive, check, store and protect items delivered.
 - 2. Store doors in conditions similar to those expected when the building is occupied. Stack leaves and panels flat and level, in supplied wrappings, on 100 x 50mm cardboard wrapping timber supports across the full width, located at 300mm from each end and not more than 800mm centers between. Inset additional spacers, if supports would cross the line of glazing beads, applied trims, or the like. Protect painted surfaces from strong sunlight, which might cause differential fading. Carefully cover with suitable coverings, to protect from dirt or damage.
 - 3. Provide suitable, labeled shelves for and nearly sort and stack ironmongery and other items supplied loose.
 - 4. Supply ironmongery templates to door makers and hold them responsible for:-
 - 5. Factory preparing doors for ironmongery, to ensure correct operation and validity of fire ratings.
 - 6. Ensuring that their sections, details reinforcement are suitable for proposed ironmongery.
 - 7. Fix door assemblies, including ironmongery, according to the most recent approved schedules, with the proper tools, to supplier's recommendations, including drilling lead holes etc., providing suitable fixings, protecting and making good finishes. Install each door leaf in the location indicated on the top edge of the door, to the correct hand with the correct face showing on each side.
 - 8. Not make any alterations to components of door assemblies which may affect fire, smoke or acoustic control ratings, including planning or trimming of any door edge, nor damage any intumescent or other seals, nor remove any permanent markings which have been affixed to indicate fire, smoke or acoustic performance. Any component so altered shall be replaced, free of charge.
 - 9. Not fixing permanent ironmongery unit all other installation works, and background finishes have been completed.
 - 10. Any damage caused to the doorsets, including door panel, frame, architrave, ironmongeries, etc., prior to substantial completion shall be replaced by the contractor at no extra time and cost to the Employer. The Architect shall retain all rights in identifying the units to be replaced.
 - 11. Upon installation and before hand-over to the employer, adequately protect wood-work at corners, surfaces, frames and ironmongery.

END OF SECTION